> **What this is.** Part 3 is gradient descent worked out by hand on paper. Part 4 (*Gradient Descent in Code*) runs the same problem in Python and must give the same numbers.
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> **How to use this file.** This is the typed answer key. The graded submission must be **handwritten**: each member writes at least one full update: **Member 3** copies Pages 1–2 (setup, chain-rule derivation, update 1); **Member 1** copies Page 3 (update 2); **Member 2** copies Pages 4–5 (update 3, summary and trend). Each person starts by restating the values they received, and signs their pages, and the pages are scanned into one PDF (see `Team_Plan.md`, section 3). Write every line, including the "small" ones — the rubric penalises skipped arithmetic.

Page 1 — Setup and Chain-Rule derivation (Member 3)

1. The model in matrix form

$$\widehat{y} = m_1 x_1 + m_2 x_2 + b \quad \Rightarrow \quad \widehat{\mathbf{y}} = X\mathbf{m} + \mathbf{b}$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, \quad \mathbf{m}^{(0)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \mathbf{b}^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Row 1 of X is data point 1: $(x_1, x_2) = (1, 3)$; row 2 is data point 2: $(4, 10)$.
- $n = 2$ data points.
- **Bias:** the equation has *one* bias b. It is written as the vector $\mathbf{b} = [b, b]^T$ only so it can be added to $X\mathbf{m}$ (one entry per data point). Both entries therefore always stay equal.
- **Learning rate:** not given in the brief, so we chose $\alpha = 0.01$. Reason: the steepest direction of this cost has curvature ≈ 127.5, and gradient descent is only stable when $\alpha < 2/127.5 \approx 0.0157$. With $\alpha = 0.1$ the cost jumps from 61 to 8089.7 after one step (diverges).

2. Cost function (Mean Squared Error)

$$J(\mathbf{m}, b) = \frac{1}{n}\sum_{i=1}^{n}\left(\widehat{y}_i - y_i\right)^2 = \frac{1}{n}\mathbf{e}^T\mathbf{e}, \qquad \mathbf{e} = \widehat{\mathbf{y}} - \mathbf{y}$$

With $n = 2$: $J = \frac{1}{2}[\left(\widehat{y}_1 - y_1\right)^2 + \left(\widehat{y}_2 - y_2\right)^2]$.

3. Derivatives with the Chain Rule

J depends on $\mathbf{m}$ and b **through** $\widehat{\mathbf{y}}$, so: $\dfrac{\partial J}{\partial \text{parameter}} = \dfrac{\partial J}{\partial \widehat{y}} \cdot \dfrac{\partial \widehat{y}}{\partial \text{parameter}}$.

Link 1 (outer): $\dfrac{\partial J}{\partial \widehat{y}_i} = \dfrac{1}{n} \cdot 2\left(\widehat{y}_i - y_i\right) \cdot 1 = \dfrac{2}{n} e_i$ (power rule: derivative of u^2 is $2u$; derivative of $\widehat{y}_i - y_i$ w.r.t. $\widehat{y}_i$ is 1).

Link 2 (inner): from $\widehat{y}_i = x_{i1}m_1 + x_{i2}m_2 + b$: $\dfrac{\partial \widehat{y}_i}{\partial m_1} = x_{i1}$, $\dfrac{\partial \widehat{y}_i}{\partial m_2} = x_{i2}$, $\dfrac{\partial \widehat{y}_i}{\partial b} = 1$.

Multiply the links and add over both data points:

$$\frac{\partial J}{\partial m_1} = \sum_i \frac{2}{n} e_i \, x_{i1} = \frac{2}{2}(e_1 \cdot 1 + e_2 \cdot 4)$$

$$\frac{\partial J}{\partial m_2} = \sum_i \frac{2}{n} e_i \, x_{i2} = \frac{2}{2}(e_1 \cdot 3 + e_2 \cdot 10)$$

$$\frac{\partial J}{\partial b} = \sum_i \frac{2}{n} e_i \cdot 1 = \frac{2}{2}(e_1 + e_2)$$

Matrix form (what we use below):

$$\frac{\partial J}{\partial \mathbf{m}} = \frac{2}{n} X^T \mathbf{e} \qquad \frac{\partial J}{\partial b} = \frac{2}{n} \mathbf{1}^T \mathbf{e} \qquad X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}, \ \frac{2}{n} = \frac{2}{2} = 1$$

Update rule: $\mathbf{m} \leftarrow \mathbf{m} - \alpha \frac{\partial J}{\partial \mathbf{m}}, \ \ b \leftarrow b - \alpha \frac{\partial J}{\partial b}.$

Page 2 — Update 1 (Member 3)

Start: $\mathbf{m} = [\,-1,\ 2\,]^T$, $b = 1$.

Step 1 – Matrix product $X\mathbf{m}$

$$X\mathbf{m} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} (1)(-1) + (3)(2) \\ (4)(-1) + (10)(2) \end{bmatrix} = \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

Step 2 – Predictions $\widehat{\mathbf{y}} = X\mathbf{m} + \mathbf{b} = \begin{bmatrix} 5+1 \\ 16+1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$

Step 3 – Error $\mathbf{e} = \widehat{\mathbf{y}} - \mathbf{y} = \begin{bmatrix} 6-5 \\ 17-6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Step 4 – Cost $J = \frac{1}{2}(1^2 + 11^2) = \frac{1}{2}(1+121) = \frac{122}{2} = 61$

Step 5 – Gradient w.r.t. m

$$X^T\mathbf{e} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}\begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} (1)(1) + (4)(11) \\ (3)(1) + (10)(11) \end{bmatrix} = \begin{bmatrix} 1+44 \\ 3+110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}, \qquad \frac{\partial J}{\partial \mathbf{m}} = 1 \times \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

Step 6 – Gradient w.r.t. b $\dfrac{\partial J}{\partial b} = 1 \times (1+11) = 12$

Step 7 – Update

$$m_1 = -1 - 0.01(45) = -1 - 0.45 = -1.45$$
$$m_2 = 2 - 0.01(113) = 2 - 1.13 = 0.87$$
$$b = 1 - 0.01(12) = 1 - 0.12 = 0.88 \quad \Rightarrow \quad \mathbf{b} = [\,0.88,\ 0.88\,]^T$$

After update 1: $\mathbf{m} = [\,-1.45,\ 0.87\,]^T$, $b = 0.88$.

Page 3 — Update 2 (Member 1)

Start: $\mathbf{m} = [\,-1.45,\ 0.87\,]^T$, $b = 0.88$.

Step 1 – Matrix product

$$X\mathbf{m} = \begin{bmatrix} (\,1\,)\,(\,-1.45\,) + (\,3\,)\,(\,0.87\,) \\ (\,4\,)\,(\,-1.45\,) + (\,10\,)\,(\,0.87\,) \end{bmatrix} = \begin{bmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{bmatrix} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$$

Step 2 – Predictions $\widehat{\mathbf{y}} = \begin{bmatrix} 1.16 + 0.88 \\ 2.90 + 0.88 \end{bmatrix} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$

Step 3 – Error $\mathbf{e} = \begin{bmatrix} 2.04 - 5 \\ 3.78 - 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$

Step 4 – Cost $J = \frac{1}{2}(\,(\,-2.96\,)^{\,2} + (\,-2.22\,)^{\,2}) = \frac{1}{2}\,(\,8.7616 + 4.9284\,) = \frac{1}{2}\,(\,13.69\,) = 6.845$

Step 5 – Gradient w.r.t. m

$$X^T\mathbf{e} = \begin{bmatrix} (\,1\,)\,(\,-2.96\,) + (\,4\,)\,(\,-2.22\,) \\ (\,3\,)\,(\,-2.96\,) + (\,10\,)\,(\,-2.22\,) \end{bmatrix} = \begin{bmatrix} -2.96 - 8.88 \\ -8.88 - 22.20 \end{bmatrix} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}, \qquad \frac{\partial J}{\partial \mathbf{m}} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

Step 6 – Gradient w.r.t. b $\dfrac{\partial J}{\partial b} = 1 \times (\,-2.96 - 2.22\,) = -5.18$

Step 7 – Update (minus a negative = plus)

$$m_1 = -1.45 - 0.01\,(\,-11.84\,) = -1.45 + 0.1184 = -1.3316$$
$$m_2 = 0.87 - 0.01\,(\,-31.08\,) = 0.87 + 0.3108 = 1.1808$$
$$b = 0.88 - 0.01\,(\,-5.18\,) = 0.88 + 0.0518 = 0.9318$$

After update 2: $\mathbf{m} = [\,-1.3316,\ 1.1808\,]^T$, $b = 0.9318$.

Page 4 — Update 3 (Member 2)

Start: $\mathbf{m} = [\,-1.3316,\ 1.1808\,]^{T}$, $b = 0.9318$.

Step 1 – Matrix product

$$X\mathbf{m} = \begin{bmatrix} (\,1\,)\ (\,-1.3316\,) + (\,3\,)\ (\,1.1808\,) \\ (\,4\,)\ (\,-1.3316\,) + (\,10\,)\ (\,1.1808\,) \end{bmatrix} = \begin{bmatrix} -1.3316 + 3.5424 \\ -5.3264 + 11.808 \end{bmatrix} = \begin{bmatrix} 2.2108 \\ 6.4816 \end{bmatrix}$$

Step 2 – Predictions $\widehat{\mathbf{y}} = \begin{bmatrix} 2.2108 + 0.9318 \\ 6.4816 + 0.9318 \end{bmatrix} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$

Step 3 – Error $\mathbf{e} = \begin{bmatrix} 3.1426 - 5 \\ 7.4134 - 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$

Step 4 – Cost $J = \frac{1}{2}(\,(\,-1.8574\,)^{\,2} + (\,1.4134\,)^{\,2}) = \frac{1}{2}\,(\,3.44993476 + 1.99769956\,) = \frac{1}{2}\,(\,5.44763432\,) = 2.72381716$

Step 5 – Gradient w.r.t. m

$$X^{T}\mathbf{e} = \begin{bmatrix} (\,1\,)\ (\,-1.8574\,) + (\,4\,)\ (\,1.4134\,) \\ (\,3\,)\ (\,-1.8574\,) + (\,10\,)\ (\,1.4134\,) \end{bmatrix} = \begin{bmatrix} -1.8574 + 5.6536 \\ -5.5722 + 14.134 \end{bmatrix} = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}, \qquad \frac{\partial J}{\partial \mathbf{m}} = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

Step 6 – Gradient w.r.t. b $\dfrac{\partial J}{\partial b} = 1 \times\ (\,-1.8574 + 1.4134\,)\ =\ -0.444$

Step 7 – Update

$$m_1 = -1.3316 - 0.01\,(\,3.7962\,) = -1.3316 - 0.037962 = -1.369562$$
$$m_2 = 1.1808 - 0.01\,(\,8.5618\,) = 1.1808 - 0.085618 = 1.095182$$
$$b = 0.9318 - 0.01\,(\,-0.444\,) = 0.9318 + 0.00444 = 0.93624$$

After update 3: $\mathbf{m} = [\,-1.369562,\ 1.095182\,]^{T}$, $b = 0.93624$.

Check of the final values: $\widehat{\mathbf{y}} = \begin{bmatrix} -1.369562 + 3.285546 + 0.93624 \\ -5.478248 + 10.95182 + 0.93624 \end{bmatrix} = \begin{bmatrix} 2.852224 \\ 6.409812 \end{bmatrix}$, $\mathbf{e} = \begin{bmatrix} -2.147776 \\ 0.409812 \end{bmatrix}$,

$J = \frac{1}{2}\,(\,4.612941746176 + 0.167945875344\,) = \frac{1}{2}\,(\,4.78088762152\,) = 2.39044381$

Page 5 — Summary and trend (Member 2)

Update	$\mathbf{m}$ before	b before	$\widehat{\mathbf{y}}$	$\mathbf{e}$	J before	$\partial J/\partial \mathbf{m}$	$\partial J/\partial b$	$\mathbf{m}$ after	b after
1	[−1, 2]	1	[6, 17]	[1, 11]	61	[45, 113]	12	[−1.45, 0.87]	0.88
2	[−1.45, 0.87]	0.88	[2.04, 3.78]	[−2.96, −2.22]	6.845	[−11.84, −31.08]	−5.18	[−1.3316, 1.1808]	0.9318
3	[−1.3316, 1.1808]	0.9318	[3.1426, 7.4134]	[−1.8574, 1.4134]	2.72381716	[3.7962, 8.5618]	−0.444	[−1.369562, 1.095182]	0.93624

Cost after the 3rd update: $J = 2.39044381$.

Trend — are m and b moving in a direction that reduces the error? Yes.

1. J falls after every update: $61 \to 6.845 \to 2.7238 \to 2.3904$ (a 96% drop in total).
2. Each parameter moves *against the sign of its gradient*: positive gradient → value decreases (update 1: m_1, m_2, b all fall); negative gradient → value increases (update 2: all rise). That is exactly "downhill".
3. The first step is the largest because the first error was the largest ($\widehat{y}_2 = 17$ vs 6). As errors shrink, gradients shrink, and steps shrink ($|\Delta m_2|$: 1.13 → 0.3108 → 0.0856).
4. m_2 zig-zags ($2 \to 0.87 \to 1.1808 \to 1.0952$) and the error of point 2 changes sign ($+11 \to -2.22 \to +1.41$): $x_2 = 10$ makes the cost very steep along m_2, so $\alpha = 0.01$ slightly overshoots, but each overshoot is smaller than the last — the zig-zag is damping, not growing.
5. The Python version (Part 4) reproduces every number above and, continued to 2000 updates, reaches $\mathbf{m} \approx [-2.2258,\ 1.0968]$, $b \approx 3.9353$, $J \approx 0$, predictions $\approx [5, 6]$.